

Student 38 **SHEKAR DHARANISH** [192472152RD](#)

2 / 2 submitted

Q1. Smart Hospital Patient Registration System

Score: 0.00 TC: 0/3 passed

CODE

```
import java.util.*;

public class Main {
    static ArrayList<String> patientIds = new ArrayList<>();
    static HashMap<String, String[]> patientMap = new HashMap<>();
    static HashMap<String, Integer> deptCount = new HashMap<>();
    static final int MAX_PATIENTS = 100;

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;

        do {
            String line = sc.nextLine().trim();
            if (line.isEmpty()) continue;
            choice = Integer.parseInt(line);

            switch (choice) {
                case 1:
                    registerPatient(sc);
                    break;
                case 2:
                    searchPatient(sc);
                    break;
                case 3:
                    displayAllPatients();
                    break;
                case 4:
                    departmentWiseCount();
                    break;
                case 5:
                    break;
                default:
                    System.out.println("Invalid Choice");
            }
        } while (choice != 5);

        sc.close();
    }

    static void registerPatient(Scanner sc) {
        if (patientIds.size() >= MAX_PATIENTS) {
            System.out.println("Maximum Patient Limit Reached.");
            return;
        }

        String id = cleanAll(sc.nextLine());
        if (patientMap.containsKey(id)) {
            System.out.println("Duplicate Patient ID.");
            return;
        }

        String nameInput = cleanAll(sc.nextLine());
        String nameNoSpaces = nameInput.replaceAll("\\s+", "");
        if (!nameNoSpaces.matches("[a-zA-Z]+")) {
            System.out.println("Invalid Name. Only alphabets allowed.");
            return;
        }

        String mobile = cleanAll(sc.nextLine());
        if (!mobile.matches("\\d{10}")) {
```

```

        System.out.println("Invalid Mobile Number. Must be 10 digits.");
        return;
    }

    String department = cleanAll(sc.nextLine()).toUpperCase();
    String doctor = cleanAll(sc.nextLine());

    String properName = toProperCase(nameInput);

    String[] patientData = { id, properName, mobile, department, doctor };

    patientIds.add(id);
    patientMap.put(id, patientData);
    deptCount.put(department, deptCount.getOrDefault(department, 0) + 1);

    System.out.println("Patient Registered Successfully.");
}

static void searchPatient(Scanner sc) {
    String id = cleanAll(sc.nextLine());

    if (patientMap.containsKey(id)) {
        String[] p = patientMap.get(id);
        System.out.println("Patient Details");
        System.out.println("ID : " + p[0]);
        System.out.println("Name : " + p[1]);
        System.out.println("Mobile : " + p[2]);
        System.out.println("Department : " + p[3]);
        System.out.println("Doctor : " + p[4]);
    } else {
        System.out.println("Patient Not Found");
    }
}

static void displayAllPatients() {
    System.out.println("-----");
    System.out.println("ID Name Mobile Department Doctor");
    System.out.println("-----");

    for (String id : patientIds) {
        String[] p = patientMap.get(id);
        System.out.println(p[0] + " " + p[1] + " " + p[2] + " " + p[3] + " " + p[4]);
    }
}

static void departmentWiseCount() {
    System.out.println("Department Count");

    TreeMap<String, Integer> sorted = new TreeMap<>(deptCount);
    for (Map.Entry<String, Integer> entry : sorted.entrySet()) {
        System.out.println(entry.getKey() + " : " + entry.getValue());
    }
}

static String cleanAll(String s) {
    if (s == null) return "";
    return s.trim().replaceAll("\\s+", " ");
}

static String toProperCase(String str) {
    str = cleanAll(str);
    if (str.isEmpty()) return str;

    String[] words = str.split("\\s+");

```

```
StringBuilder result = new StringBuilder();

for (String word : words) {
    if (word.length() > 0) {
        result.append(Character.toUpperCase(word.charAt(0)))
                .append(word.substring(1).toLowerCase())
                .append(" ");
    }
}
return result.toString().trim();
}
```

INPUT

```
1
P101
Rahul Kumar
9876543210
Cardiology
Dr.Arun
1
P102
Priya
9876543211
Ortho
Dr.Meena
3
4
2
P101
5
```

OUTPUT

```
Compilation Error:
C:\Windows\TEMP\sandbox_192472152RD_6a6b7b46628934.39902480\Main.java:36: error: variable choice might not have been initiali
zed
        } while (choice != 5);
                ^
1 error
```

Q2. Online Library Book Management System**CODE**

```
import java.util.*;

class BookManagement {
    private String bookId;
    private String title;
    private String author;
    private String category;
    private String publisher;

    public BookManagement(String bookId, String title, String author, String category, String publisher) {
        this.bookId = bookId.trim();
        this.title = toTitleCase(title.trim());
        this.author = author.trim();
        this.category = category.trim().toUpperCase();
        this.publisher = publisher.trim();
    }

    public String getBookId() { return bookId; }
    public String getTitle() { return title; }
    public String getAuthor() { return author; }
    public String getCategory() { return category; }
    public String getPublisher() { return publisher; }

    private String toTitleCase(String input) {
        if (input == null || input.isEmpty()) return input;
        String[] words = input.split("\\s+");
        StringBuilder sb = new StringBuilder();
        for (int i = 0; i < words.length; i++) {
            if (!words[i].isEmpty()) {
                sb.append(Character.toUpperCase(words[i].charAt(0)));
                if (words[i].length() > 1) {
                    sb.append(words[i].substring(1).toLowerCase());
                }
                if (i < words.length - 1) {
                    sb.append(" ");
                }
            }
        }
        return sb.toString();
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        List<BookManagement> bookList = new ArrayList<>();
        Map<String, BookManagement> bookMap = new HashMap<>();
        Set<String> authorSet = new LinkedHashSet<>();
        Map<String, Integer> categoryCountMap = new LinkedHashMap<>();

        while (sc.hasNextLine()) {
            String line = sc.nextLine().trim();
            if (line.isEmpty()) continue;

            int choice;
            try {
                choice = Integer.parseInt(line);
            } catch (NumberFormatException e) {
                continue;
            }

            switch (choice) {
                case 1: {
                    if (!sc.hasNextLine()) break;
```

```

String id = sc.nextLine().trim();
if (!sc.hasNextLine()) break;
String title = sc.nextLine();
if (!sc.hasNextLine()) break;
String author = sc.nextLine();
if (!sc.hasNextLine()) break;
String category = sc.nextLine();
if (!sc.hasNextLine()) break;
String publisher = sc.nextLine();

if (bookMap.containsKey(id)) {
    break;
}

BookManagement book = new BookManagement(id, title, author, category, publisher);
bookList.add(book);
bookMap.put(id, book);
authorSet.add(book.getAuthor());
categoryCountMap.put(book.getCategory(), categoryCountMap.getOrDefault(book.getCategory(), 0) + 1);

System.out.println("Book Added Successfully.");
break;
}

case 2: {
    if (!sc.hasNextLine()) break;
    String searchId = sc.nextLine().trim();
    BookManagement book = bookMap.get(searchId);

    if (book == null) {
        System.out.println("Book Not Available");
    } else {
        System.out.println("Book Details");
        System.out.println("Book ID : " + book.getBookId());
        System.out.println("Title : " + book.getTitle());
        System.out.println("Author : " + book.getAuthor());
        System.out.println("Category : " + book.getCategory());
        System.out.println("Publisher : " + book.getPublisher());
    }
    break;
}

case 3: {
    System.out.println("-----");
    System.out.println("Book ID    Title                Author");
    System.out.println("-----");
    for (BookManagement b : bookList) {
        System.out.println(b.getBookId() + "    " + b.getTitle() + "    " + b.getAuthor());
    }
    break;
}

case 4: {
    System.out.println("Unique Authors");
    for (String author : authorSet) {
        System.out.println(author);
    }
    break;
}

case 5: {
    System.out.println("Category-wise Count");
    for (Map.Entry<String, Integer> entry : categoryCountMap.entrySet()) {
        System.out.println(entry.getKey() + " : " + entry.getValue());
    }
}

```

```
        break;
    }

    case 6:
        sc.close();
        return;

    default:
        break;
    }
}
sc.close();
}
```

INPUT

```
1
B101
Java Programming
Herbert Schildt
Programming
McGraw Hill
1
B102
Data Structures
Mark Allen
Computer Science
Pearson
2
B101
3
4
5
6
```

OUTPUT

(no output)

Student 39 **Vijay V** 192572123RD

0 / 2 submitted

Q1. Smart Hospital Patient Registration System

— Not Submitted —

Q2. Online Library Book Management System

— Not Submitted —